Supplementary for Coordinate Descent

Motivation

Gradient-based algorithms we have studied require FULL computations of x, ∇f and $P_{\alpha_k g}$:

- **Gradient descent method** for $\min\limits_{x\in\mathbb{R}^n} \; f(x)$

$$f : \mathbb{R}^n \to \mathbb{R} \text{ convex differentiable}$$

$$x^{(k+1)} = x^{(k)} - \alpha_k \nabla f(x^{(k)})$$

- **(Accelerated) proximal gradient method** for $\min\limits_{x\in\mathbb{R}^n} \; f(x) + g(x)$

$$f : \mathbb{R}^n \to \mathbb{R} \text{ convex differentiable, } \nabla f \; L\text{-Lipschitz continuous}$$

$$g : \mathbb{R}^n \to (-\infty, +\infty] \text{ closed proper convex } (g = \lambda\|\cdot\|_1,\; g = \delta_C)$$

$$x^{(k+1)} = P_{\alpha_k g}\left(x^{(k)} - \alpha_k \nabla f(x^{(k)})\right) \; (+ \text{ acceleration})$$

These may be difficult to implement when problem is complex or in large-scale !?

A Counter Example

Question: is the **coordinate descent (CD)** method always convergent?

[1] Michael J.D. Powell. "On search directions for minimization algorithms." Mathematical Programming, 4: 193–201, 1973.

A Counter Example

Question: is the **coordinate descent (CD)** method always convergent? **NO** !

[1] Michael J.D. Powell. "On search directions for minimization algorithms." Mathematical Programming, 4: 193–201, 1973.

A Counter Example

Question: is the **coordinate descent (CD)** method always convergent? **NO** !

Consider a popular example (Powell '73)[1] as follows:

$$\min\ -x_1x_2 - x_2x_3 - x_3x_1 + \sum_{i=1}^{3}[(x_i - 1)_+^2 + (-x_i - 1)_+^2].$$

Note that, fixing (x_2, x_3) and optimizing over x_1 yields that following solution

$$x_1 \in \begin{cases} (1 + 0.5|x_2 + x_3|)\,\mathrm{sign}(x_2 + x_3), & \text{if } x_2 + x_3 \neq 0, \\ [-1,\ 1], & \text{otherwise.} \end{cases}$$

A similar solution can be obtained for x_2 and x_3 as well.

[1] Michael J.D. Powell. "On search directions for minimization algorithms." Mathematical Programming, 4: 193–201, 1973.

A Counter Example (cont.)

Start Coordinate Descent Method from $(-1-\varepsilon, 1+\frac{1}{2}\varepsilon, -1-\frac{1}{4}\varepsilon)$ with some $\varepsilon > 0$, then minimizing F over x_1, x_2, x_3, x_1, x_2, x_3, ... produces

$$\xrightarrow{x_1} (1+\tfrac{1}{8}\varepsilon, 1+\tfrac{1}{2}\varepsilon, -1-\tfrac{1}{4}\varepsilon) \qquad \xrightarrow{x_2} (1+\tfrac{1}{8}\varepsilon, -1-\tfrac{1}{16}\varepsilon, -1-\tfrac{1}{4}\varepsilon)$$

$$\xrightarrow{x_3} (1+\tfrac{1}{8}\varepsilon, -1-\tfrac{1}{16}\varepsilon, 1+\tfrac{1}{32}\varepsilon) \qquad \xrightarrow{x_1} (-1-\tfrac{1}{64}\varepsilon, -1-\tfrac{1}{16}\varepsilon, 1+\tfrac{1}{32}\varepsilon)$$

$$\xrightarrow{x_2} (-1-\tfrac{1}{64}\varepsilon, 1+\tfrac{1}{128}\varepsilon, 1+\tfrac{1}{32}\varepsilon) \qquad \xrightarrow{x_3} (-1-\tfrac{1}{64}\varepsilon, 1+\tfrac{1}{128}\varepsilon, -1-\tfrac{1}{256}\varepsilon)$$

$$\ldots \qquad \ldots$$

So, the iterates will be cycling around 6 points $(1, 1, -1)$, $(1, -1, -1)$, $(1, -1, 1)$, $(-1, -1, 1)$, $(-1, 1, 1)$, $(-1, 1, -1)$.

Coordinate-wise Minimizer

Definition 1 (Coordinate-wise minimizer)

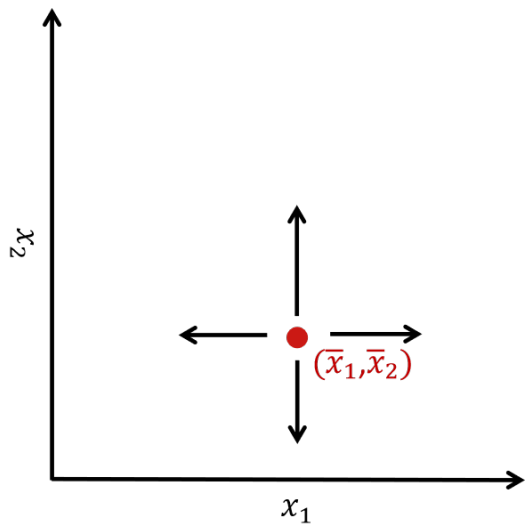
For any $f : \mathbb{R}^n \to (-\infty, +\infty]$, we say $\bar{x}$ is a coordinate-wise minimizer of f if $\bar{x} \in \mathrm{dom} f$ and

$$f(\bar{x} + de_i) \geq f(\bar{x}) \quad \forall\, i \in [n],\, d \in \mathbb{R} \tag{1}$$

where $e_i = (0, \ldots, 1, \ldots, 0)^T \in \mathbb{R}^n$ is the i-th standard basis vector.

- When $n = 2$, (1) is

$$f(\bar{x}_1 + d, \bar{x}_2) \geq f(\bar{x}_1, \bar{x}_2)$$
$$f(\bar{x}_1, \bar{x}_2 + d) \geq f(\bar{x}_1, \bar{x}_2) \quad \forall\, d \in \mathbb{R}$$



- When $n = 3$, (1) is

$$f(\bar{x}_1 + d, \bar{x}_2, \bar{x}_3) \geq f(\bar{x}_1, \bar{x}_2, \bar{x}_3)$$
$$f(\bar{x}_1, \bar{x}_2 + d, \bar{x}_3) \geq f(\bar{x}_1, \bar{x}_2, \bar{x}_3)$$
$$f(\bar{x}_1, \bar{x}_2, \bar{x}_3 + d) \geq f(\bar{x}_1, \bar{x}_2, \bar{x}_3) \quad \forall\, d \in \mathbb{R}$$

Coordinate-wise Minimizer (cont.)

- For general n, (1) is equivalent to

$$\bar{x}_i \in \arg\min_{x_i} f(\bar{x}_1, \ldots, \bar{x}_{i-1}, x_i, \bar{x}_{i+1}, \ldots, \bar{x}_n), \quad \forall\, i \in [n].$$

Coordinate-wise Minimizer (cont.)

- For general n, (1) is equivalent to

$$\bar{x}_i \in \arg\min_{x_i} f(\bar{x}_1, \ldots, \bar{x}_{i-1}, x_i, \bar{x}_{i+1}, \ldots, \bar{x}_n), \quad \forall\, i \in [n].$$

- **Question**: a coordinate-wise minimizer $\overset{?}{\Longrightarrow}$ a global minimizer

Example 1

Verify that $(\bar{x}_1, \bar{x}_2) = (-3, -3)$ is a coordinate-wise minimizer[1] of

$$f(x_1, x_2) = x_1^2 + x_2^2 + 20|x_1 - x_2|$$

Solution.

[1] The coordinate-wise minimizer may not be unique.

Example 1

Verify that $(\bar{x}_1, \bar{x}_2) = (-3, -3)$ is a coordinate-wise minimizer[1] of

$$f(x_1, x_2) = x_1^2 + x_2^2 + 20|x_1 - x_2|$$

Solution. We need to verify that

$$\bar{x}_1 \in \arg\min_{x_1} \; f(x_1, \bar{x}_2) = x_1^2 + 20|x_1 + 3| + 9 \qquad (*)$$

$$\bar{x}_2 \in \arg\min_{x_2} \; f(\bar{x}_1, x_2) = x_2^2 + 20|x_2 + 3| + 9$$

Namely, -3 is a global minimizer of $\min_x \; x^2 + 20|x + 3|$

[1] The coordinate-wise minimizer may not be unique.

Example 1

Verify that $(\bar{x}_1, \bar{x}_2) = (-3, -3)$ is a coordinate-wise minimizer[1] of

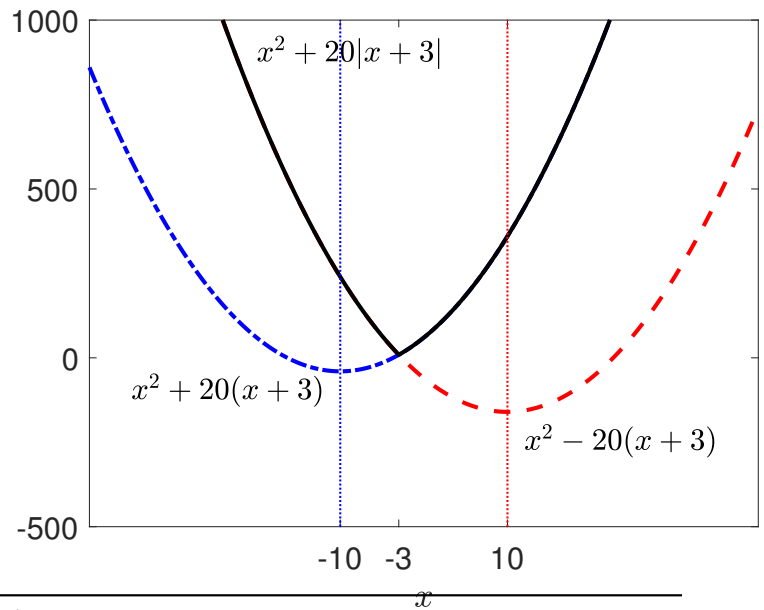
$$f(x_1, x_2) = x_1^2 + x_2^2 + 20|x_1 - x_2|$$

Solution. We need to verify that

$$\bar{x}_1 \in \arg\min_{x_1} \; f(x_1, \bar{x}_2) = x_1^2 + 20|x_1 + 3| + 9 \qquad (*)$$

$$\bar{x}_2 \in \arg\min_{x_2} \; f(\bar{x}_1, x_2) = x_2^2 + 20|x_2 + 3| + 9$$

Namely, -3 is a global minimizer of $\min_x \; x^2 + 20|x + 3|$



Method 1: verify graphically

$$\begin{cases} x^2 + 20(x+3), & \text{if } x \geq -3 \\ x^2 - 20(x+3), & \text{if } x < -3 \end{cases}$$

[1] The coordinate-wise minimizer may not be unique.

Example 1 (cont.)

Verify that $(\bar{x}_1, \bar{x}_2) = (-3, -3)$ is a coordinate-wise minimizer of

$$f(x_1, x_2) = x_1^2 + x_2^2 + 20|x_1 - x_2|$$

Solution. We need to verify that

$$\bar{x}_1 \in \arg\min_{x_1} \; f(x_1, \bar{x}_2) = x_1^2 + 20|x_1 + 3| + 9 \qquad (*)$$
$$\bar{x}_2 \in \arg\min_{x_2} \; f(\bar{x}_1, x_2) = x_2^2 + 20|x_2 + 3| + 9$$

Method 2: Recall that $x^* \in \arg\min f(x) \iff 0 \in \partial f(x^*)$; If $g(x) = |x|$, $x \in \mathbb{R}$, then $\partial g(0) = [-1, 1]$.

(*) holds due to that $0 \in 2\bar{x}_1 + 20\,\partial g(\bar{x}_1 + 3) = -6 + [-20, 20]$.

Example 1 (cont.)

Verify that $(\bar{x}_1, \bar{x}_2) = (-3, -3)$ is a coordinate-wise minimizer of

$$f(x_1, x_2) = x_1^2 + x_2^2 + 20|x_1 - x_2|$$

Solution. We need to verify that

$$\bar{x}_1 \in \arg\min_{x_1} \; f(x_1, \bar{x}_2) = x_1^2 + 20|x_1 + 3| + 9 \qquad (*)$$
$$\bar{x}_2 \in \arg\min_{x_2} \; f(\bar{x}_1, x_2) = x_2^2 + 20|x_2 + 3| + 9$$

Method 2: Recall that $x^* \in \arg\min f(x) \iff 0 \in \partial f(x^*)$; If $g(x) = |x|$, $x \in \mathbb{R}$, then $\partial g(0) = [-1, 1]$.

(*) holds due to that $0 \in 2\bar{x}_1 + 20\,\partial g(\bar{x}_1 + 3) = -6 + [-20, 20]$.

However, $(-3, -3)$ is not a global minimizer !!

In fact, it is clear that a global minimizer is $(0, 0)$.

Example 2

Verify that $(\bar{x}_1, \bar{x}_2) = (0, 0)$ is a coordinate-wise minimizer of

$$f(x_1, x_2) = (x_1 - x_2)^2 + |x_1| + |x_2|$$

Solution. We need to verify that

$$\bar{x}_1 \in \arg\min_{x_1} \; f(x_1, \bar{x}_2) = x_1^2 + |x_1|$$
$$\bar{x}_2 \in \arg\min_{x_2} \; f(\bar{x}_1, x_2) = x_2^2 + |x_2|$$

0 is a global minimizer of $\min_x \; x^2 + |x|$. ($x^2 + |x| \geq 0$ for any x)

Example 2

Verify that $(\bar{x}_1, \bar{x}_2) = (0, 0)$ is a coordinate-wise minimizer of

$$f(x_1, x_2) = (x_1 - x_2)^2 + |x_1| + |x_2|$$

Solution. We need to verify that

$$\bar{x}_1 \in \arg\min_{x_1} \; f(x_1, \bar{x}_2) = x_1^2 + |x_1|$$
$$\bar{x}_2 \in \arg\min_{x_2} \; f(\bar{x}_1, x_2) = x_2^2 + |x_2|$$

0 is a global minimizer of $\min_x \; x^2 + |x|$. ($x^2 + |x| \geq 0$ for any x)

For this example, $(0, 0)$ is also a global minimizer.

Coordinate-wise Minimizer: differentiable

?? a coordinate-wise minimizer $\overset{\text{differentiable}}{\Longrightarrow}$ a global minimizer

Claim 1

A coordinate-wise minimizer $\bar{x}$ of a convex function f is a global minimizer of f whenever f is differentiable at $\bar{x}$.

Proof: Since f is differentiable at $\bar{x}$,

$$\bar{x}_i \in \arg\min_{x_i} f(\bar{x}_1, \ldots, \bar{x}_{i-1}, x_i, \bar{x}_{i+1}, \ldots, \bar{x}_n), \quad \forall\, i \in [n].$$

implies that

$$\nabla_i f(\bar{x}) = \frac{\partial}{\partial x_i} f(\bar{x}) = 0, \quad \forall\, i \in [n].$$

Thus, $\nabla f(\bar{x}) = (\nabla_1 f(\bar{x}), \ldots, \nabla_n f(\bar{x})) = 0$, $\bar{x}$ is a global minimizer of f.

Coordinate-wise Minimizer: differentiable

?? a coordinate-wise minimizer $\overset{\text{differentiable}}{\Longrightarrow}$ a global minimizer

Claim 1

A coordinate-wise minimizer $\bar{x}$ of a convex function f is a global minimizer of f whenever f is differentiable at $\bar{x}$.

Proof: Since f is differentiable at $\bar{x}$,

$$\bar{x}_i \in \arg\min_{x_i} f(\bar{x}_1, \ldots, \bar{x}_{i-1}, x_i, \bar{x}_{i+1}, \ldots, \bar{x}_n), \quad \forall\, i \in [n].$$

implies that

$$\nabla_i f(\bar{x}) = \frac{\partial}{\partial x_i} f(\bar{x}) = 0, \quad \forall\, i \in [n].$$

Thus, $\nabla f(\bar{x}) = (\nabla_1 f(\bar{x}), \ldots, \nabla_n f(\bar{x})) = 0$, $\bar{x}$ is a global minimizer of f.

How about non-differentiable function?

Coordinate-wise Minimizer: non-differentiable

?? a coordinate-wise minimizer $\overset{\text{non-differentiable}}{\Longrightarrow}$ a global minimizer

Claim 2

A coordinate-wise minimizer $\bar{x}$ of a convex function f is not necessarily a global minimizer of f when f is not differentiable at $\bar{x}$.

Coordinate-wise Minimizer: non-differentiable

?? a coordinate-wise minimizer $\overset{\text{non-differentiable}}{\Longrightarrow}$ a global minimizer

Claim 2

A coordinate-wise minimizer $\bar{x}$ of a convex function f is not necessarily a global minimizer of f when f is not differentiable at $\bar{x}$.

Example: $f : \mathbb{R}^2 \to \mathbb{R}$ is convex but not differentiable when $x_1 = x_2$

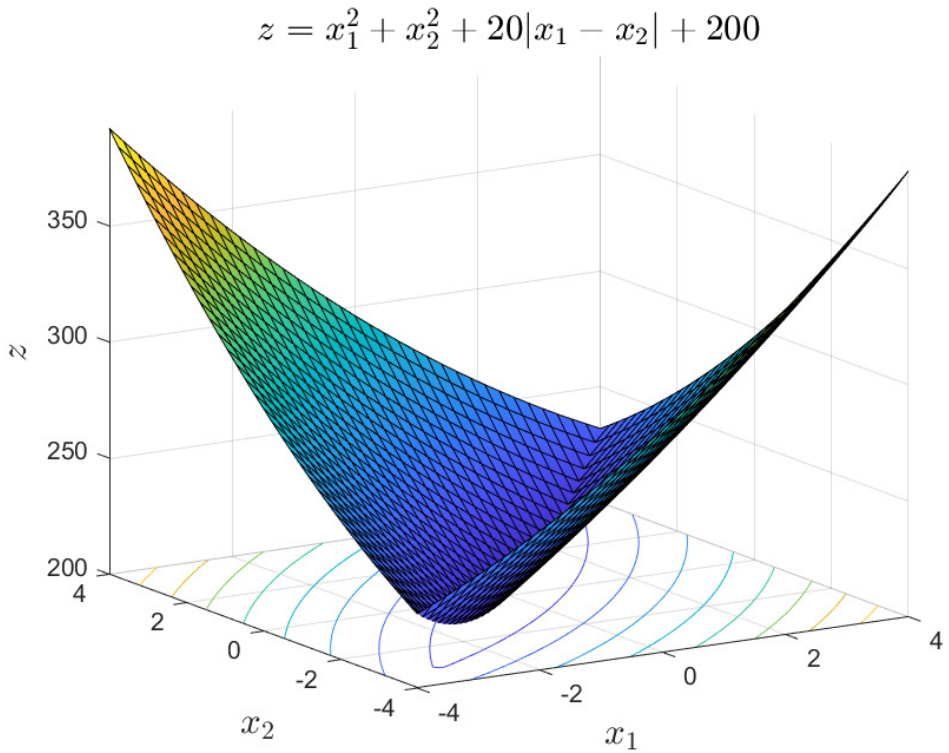
$$f(x_1, x_2) = \begin{cases} (x_1 + 10)^2 + (x_2 - 10)^2, & \text{if } x_1 \geq x_2 \\ (x_1 - 10)^2 + (x_2 + 10)^2, & \text{if } x_1 < x_2 \end{cases}$$

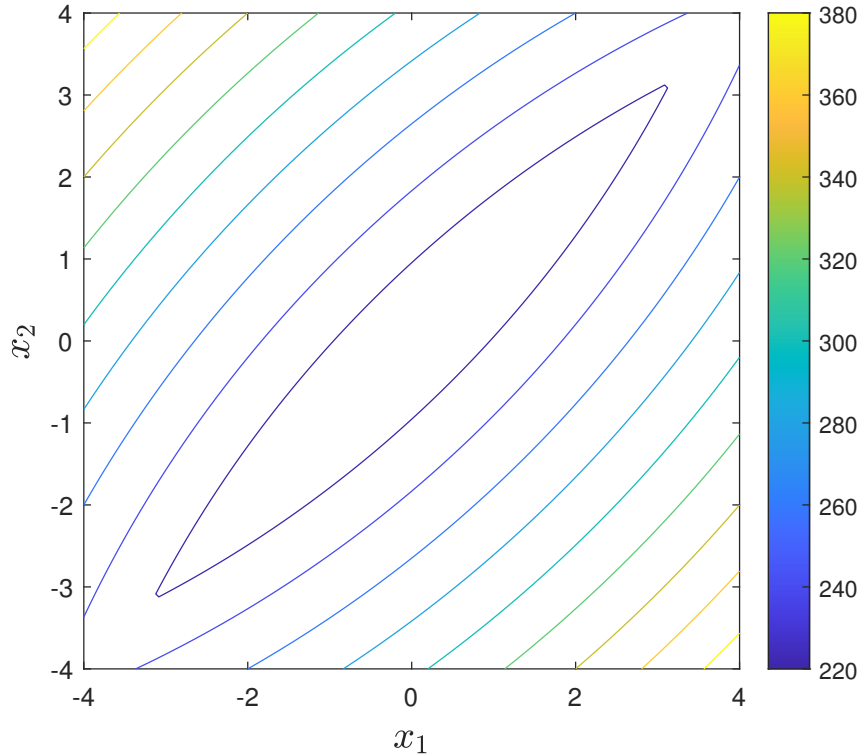
$$= x_1^2 + x_2^2 + 20|x_1 - x_2| + 200$$

- $(3, 3)$ is a coordinate-wise minimizer of f (In fact, any point (c, c), $|c| \leq 10$ is a coordinate-wise minimizer).
- The global minimizer of f is $(0, 0)$.

Coordinate-wise Minimizer: non-differentiable (cont.)

Plot and contour plot of $f(x_1, x_2) = x_1^2 + x_2^2 + 20|x_1 - x_2| + 200$





Coordinate-wise Minimizer: special non-differentiable

a coordinate-wise minimizer $\overset{\text{non-differentiable}}{\underset{\text{but special}}{\Longrightarrow}}$ a global minimizer

Claim 3

Consider the problem where the non-differentiable part can be decomposed into a sum of functions over each coordinate

$$\min_{x\in\mathbb{R}^n} \quad F(x) := f(x) + \underbrace{\textstyle\sum_{i=1}^n r_i(x_i)}_{\text{separable}}$$

$f : \mathbb{R}^n \to \mathbb{R}$ convex differentiable

each $r_i : \mathbb{R}^n \to (-\infty, +\infty]$ closed proper convex

A coordinate-wise minimizer of F is a global minimizer of F.

Proof: Denote $r(x) := \sum_{i=1}^n r_i(x_i)$.

$$\bar{x}_i \in \arg\min_{x_i} \; f(\bar{x}_1, \ldots, \bar{x}_{i-1}, x_i, \bar{x}_{i+1}, \ldots, \bar{x}_n) + r_i(x_i) \quad \forall\, i$$

$$\iff \quad 0 \in \nabla_i f(\bar{x}) + \partial r_i(\bar{x}_i) \quad \forall\, i \iff 0 \in \nabla f(\bar{x}) + \partial r(\bar{x})$$

Target Problem

Target problem

$$\min_{x\in\mathbb{R}^n} \quad F(x) := f(x) + \sum_{i=1}^{n} r_i(x_i)$$

$f : \mathbb{R}^n \to \mathbb{R}$ convex differentiable

each $r_i : \mathbb{R}^n \to (-\infty, +\infty]$ closed proper convex

Why this problem?

- Roughly speaking, the separable structure can facilitate the application of a coordinate descent method, which searchs for a coordinate-wise minimizer
- In general, a coordinate-wise minimizer is not necessarily a global minimizer
- For the target problem, a coordinate-wise minimizer is indeed a global minimizer

Coordinate Descent Method

Algorithm (Coordinate descent method)

Choose $x^{(0)} \in \operatorname{dom} F$. Set $k \leftarrow 0$

repeat until convergence

$$
\begin{aligned}
x_1^{(k+1)} &\leftarrow \arg\min_{x_1} f(x_1, x_2^{(k)}, x_3^{(k)}, \ldots, x_n^{(k)}) + r_1(x_1) \\
x_2^{(k+1)} &\leftarrow \arg\min_{x_2} f(x_1^{(k+1)}, x_2, x_3^{(k)}, \ldots, x_n^{(k)}) + r_2(x_2) \\
x_3^{(k+1)} &\leftarrow \arg\min_{x_3} f(x_1^{(k+1)}, x_2^{(k+1)}, x_3, \ldots, x_n^{(k)}) + r_3(x_3) \\
&\vdots \\
x_n^{(k+1)} &\leftarrow \arg\min_{x_n} f(x_1^{(k+1)}, x_2^{(k+1)}, x_3^{(k+1)}, \ldots, x_n) + r_n(x_n)
\end{aligned}
$$

$k \leftarrow k + 1$

end(repeat)

Toy Example

Apply coordinate descent method for

$$\min_{x=(x_1,x_2)\in\mathbb{R}^2} \quad f(x_1,x_2) = (x_1-x_2)^2 + |x_1| + |x_2|$$

with initial point $x^{(0)} = (6,6)$.

Iterations: for $k = 0,1,2,\ldots,$

$$x_1^{(k+1)} \in \arg\min_{x_1} \; f(x_1, x_2^{(k)}) = \arg\min_{x_1} \; \left(x_1 - x_2^{(k)}\right)^2 + |x_1|$$

$$x_2^{(k+1)} \in \arg\min_{x_2} \; f(x_1^{(k+1)}, x_2) = \arg\min_{x_2} \; \left(x_2 - x_1^{(k+1)}\right)^2 + |x_2|$$

By computing the proximal mapping of $|\cdot|$,

$$x_1^{(k+1)} = \arg\min_{x_1} \; \frac{1}{2}|x_1| + \frac{1}{2}\left(x_1 - x_2^{(k)}\right)^2 = P_{0.5|\cdot|}\left(x_2^{(k)}\right) = S_{0.5}\left(x_2^{(k)}\right)$$

$$x_2^{(k+1)} = S_{0.5}\left(x_1^{(k+1)}\right)$$

Toy Example (cont.)

Apply coordinate descent method for

$$\min_{x=(x_1,x_2)\in\mathbb{R}^2} \quad f(x_1,x_2) = (x_1 - x_2)^2 + |x_1| + |x_2|$$

with initial point $x^{(0)} = (6,6)$. Compute $x^{(1)}$ and $x^{(2)}$.

Iterations:

$$x_1^{(k+1)} = S_{0.5}\left(x_2^{(k)}\right), \qquad x_2^{(k+1)} = S_{0.5}\left(x_1^{(k+1)}\right)$$

$k = 0$. $x^{(1)} = (5.5, 5)$

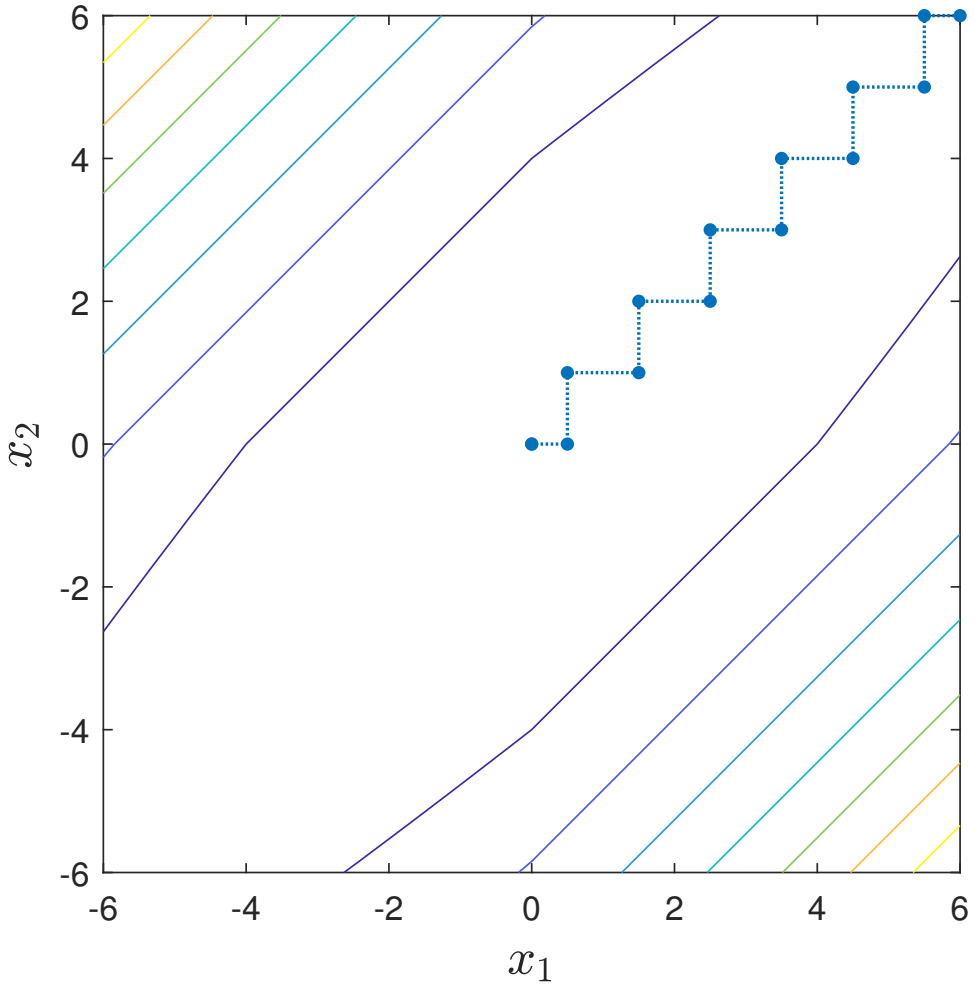
$$x_1^{(1)} = S_{0.5}\left(x_2^{(0)}\right) = S_{0.5}(6) = 5.5,\ x_2^{(1)} = S_{0.5}\left(x_1^{(1)}\right) = S_{0.5}(5.5) = 5$$

$k = 1$. $x^{(2)} = (4.5, 4)$

$$x_1^{(2)} = S_{0.5}\left(x_2^{(1)}\right) = S_{0.5}(5) = 4.5,\ x_2^{(2)} = S_{0.5}\left(x_1^{(2)}\right) = S_{0.5}(4.5) = 4$$

Toy Example (cont.)

Contour plot of $f(x_1, x_2) = (x_1 - x_2)^2 + |x_1| + |x_2|$ and iterates $x^{(k)}$.



Applications

Application 1: Linear Regression

Consider

$$\min_{\boldsymbol{x}\in\mathbb{R}^p} \quad f(\boldsymbol{x}) := \frac{1}{2}\|A\boldsymbol{x} - \boldsymbol{b}\|^2,$$

where $A \in \mathbb{R}^{m\times n}$ and $\boldsymbol{b} \in \mathbb{R}^m$.

Application 1: Linear Regression

Consider

$$\min_{\boldsymbol{x}\in\mathbb{R}^p} \quad f(\boldsymbol{x}) := \frac{1}{2}\|A\boldsymbol{x} - \boldsymbol{b}\|^2,$$

where $A \in \mathbb{R}^{m\times n}$ and $\boldsymbol{b} \in \mathbb{R}^m$.

First, it is natural to apply the gradient descent (GD) method:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - \alpha\nabla f(\boldsymbol{x}^k) = \boldsymbol{x}^k - \alpha A^\top(A\boldsymbol{x}^k - \boldsymbol{b}).$$

Application 1: Linear Regression

Consider

$$\min_{\boldsymbol{x}\in\mathbb{R}^p} \quad f(\boldsymbol{x}) := \frac{1}{2}\|A\boldsymbol{x}-\boldsymbol{b}\|^2,$$

where $A \in \mathbb{R}^{m\times n}$ and $\boldsymbol{b} \in \mathbb{R}^m$.

First, it is natural to apply the gradient descent (GD) method:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - \alpha\nabla f(\boldsymbol{x}^k) = \boldsymbol{x}^k - \alpha A^\top(A\boldsymbol{x}^k - \boldsymbol{b}).$$

We can also apply the coordinate descent (CD) method. Indeed, consider minimizing over x_i with all x_j, $j \neq i$ fixed. Note that

$$A\boldsymbol{x} = A_{\cdot 1}x_1 + A_{\cdot 2}x_2 + \cdots + A_{\cdot p}x_p = A_{\cdot i}x_i + A_{-i}\boldsymbol{x}_{-i}$$

where $A = [A_{\cdot 1}\cdots A_{\cdot(i-1)}\ A_{\cdot i}\ A_{\cdot(i+1)}\cdots A_{\cdot p}]$

$A_{-i} = [A_{\cdot 1}\cdots A_{\cdot(i-1)}\quad A_{\cdot(i+1)}\cdots A_{\cdot p}]$ delete i-th column

$\boldsymbol{x}_{-i} = [x_1\cdots x_{i-1}\ x_{i+1}\cdots x_p]^\top$ delete i-th entry

Therefore, $f(\boldsymbol{x}) = \frac{1}{2}\|A_{\cdot i}x_i + A_{-i}\boldsymbol{x}_{-i} - \boldsymbol{b}\|^2$ and we set

$$0 = \nabla_i f(\boldsymbol{x}) = A_{\cdot i}^\top(A_{\cdot i}x_i + A_{-i}\boldsymbol{x}_{-i} - \boldsymbol{b}) \quad \Rightarrow \quad x_i = \frac{A_{\cdot i}^\top(\boldsymbol{b} - A_{-i}\boldsymbol{x}_{-i})}{\|A_{\cdot i}\|^2}$$

Application 1: Linear Regression (cont.)

Algorithm (Coordinate Descent Method for Linear Regression)

Initialize $\boldsymbol{x}$.
repeat until convergence
  for $i = 1, \ldots, n$

$$x_i \quad \leftarrow \quad \frac{A_{\cdot i}^{\top}(\boldsymbol{b} - A_{-i}\boldsymbol{x}_{-i})}{\|A_{\cdot i}\|^2}$$

  end(for)
end(repeat)

Note: update of x_i can also be written as $x_i \quad \leftarrow \quad x_i - \frac{A_{\cdot i}^{\top}(A\boldsymbol{x} - \boldsymbol{b})}{\|A_{\cdot i}\|^2}$

Synthetic Data for Linear Regression

- Generate an $m \times n$ random matrix A, each entry follows a standard normal distribution $A_{ij} \sim N(0, 1)$
- Generate a sparse $\boldsymbol{x}_{\text{true}} \in \mathbb{R}^n$
- The response vector

$$\boldsymbol{b} = A\boldsymbol{x}_{\text{true}} + 0.1\epsilon$$

where $\epsilon_i \sim N(0, 1)$, $i \in [n]$ is the Gaussian noise.

GD vs CD

Coordinate descent: $x_i \leftarrow x_i - \dfrac{A_{\cdot i}^\top (A\boldsymbol{x} - \boldsymbol{b})}{\|A_{\cdot i}\|^2}$, $i = 1, \ldots, p$

Gradient descent: $x_i \leftarrow x_i - \alpha A_{\cdot i}^\top (A\boldsymbol{x} - \boldsymbol{b})$, $i = 1, \ldots, p$

GD vs CD

Coordinate descent: $x_i \leftarrow x_i - \dfrac{A_{\cdot i}^\top (A\boldsymbol{x} - \boldsymbol{b})}{\|A_{\cdot i}\|^2}$, $i = 1, \ldots, p$

Gradient descent: $x_i \leftarrow x_i - \alpha A_{\cdot i}^\top (A\boldsymbol{x} - \boldsymbol{b})$, $i = 1, \ldots, p$

From numerical results, we see

- Coordinate descent is often faster than gradient descent with a constant step size for linear regression. **Why?**

GD vs CD

Coordinate descent: $x_i \leftarrow x_i - \dfrac{A_{\cdot i}^\top (A\boldsymbol{x} - \boldsymbol{b})}{\|A_{\cdot i}\|^2}$, $i = 1, \ldots, p$

Gradient descent: $x_i \leftarrow x_i - \alpha A_{\cdot i}^\top (A\boldsymbol{x} - \boldsymbol{b})$, $i = 1, \ldots, p$

From numerical results, we see

- Coordinate descent is often faster than gradient descent with a constant step size for linear regression. **Why?**
- Coordinate descent is "parameter-free", but cannot be performed in parallel. Gradient descent needs to choose step size α, it can be performed in parallel.

GD vs CD

Coordinate descent: $x_i \leftarrow x_i - \dfrac{A_{\cdot i}^\top (A\boldsymbol{x} - \boldsymbol{b})}{\|A_{\cdot i}\|^2}$, $i = 1, \ldots, p$

Gradient descent: $x_i \leftarrow x_i - \alpha A_{\cdot i}^\top (A\boldsymbol{x} - \boldsymbol{b})$, $i = 1, \ldots, p$

From numerical results, we see

- Coordinate descent is often faster than gradient descent with a constant step size for linear regression. **Why?**
- Coordinate descent is "parameter-free", but cannot be performed in parallel. Gradient descent needs to choose step size α, it can be performed in parallel.

Further,

- Plot the first 50 entries of $\boldsymbol{x}_{\text{true}}$ and the computed solution $\boldsymbol{x}$.
- Many entries of $\boldsymbol{x}_{\text{true}}$ is zero. However, the estimated $\boldsymbol{x}$ from linear regression may not be sparse.

Application 2: Lasso

Consider

$$\min_{x \in \mathbb{R}^n} \quad \frac{1}{2}\|A\boldsymbol{x} - \boldsymbol{b}\|^2 + \lambda\|\boldsymbol{x}\|_1,$$

where $A \in \mathbb{R}^{m \times n}$, $\boldsymbol{b} \in \mathbb{R}^m$, $\lambda > 0$. The non-differentiable term is separable: $\lambda\|\boldsymbol{x}\|_1 = \sum_{i=1}^{p} \lambda|x_i|$. Let $g(\boldsymbol{x}) := \frac{1}{2}\|A\boldsymbol{x} - \boldsymbol{b}\|^2$.

Application 2: Lasso

Consider

$$\min_{x\in\mathbb{R}^n} \quad \frac{1}{2}\|A\boldsymbol{x}-\boldsymbol{b}\|^2+\lambda\|\boldsymbol{x}\|_1,$$

where $A\in\mathbb{R}^{m\times n}$, $\boldsymbol{b}\in\mathbb{R}^m$, $\lambda>0$. The non-differentiable term is separable: $\lambda\|\boldsymbol{x}\|_1=\sum_{i=1}^{p}\lambda|x_i|$. Let $g(\boldsymbol{x}):=\frac{1}{2}\|A\boldsymbol{x}-\boldsymbol{b}\|^2$.

First, we can apply the proximal gradient (PG) method:

$$\boldsymbol{x}^{k+1}=\texttt{prox}_{\alpha\lambda\|\cdot\|_1}\left(\boldsymbol{x}^k-\alpha\nabla g(\boldsymbol{x}^k)\right)=\texttt{prox}_{\alpha\lambda\|\cdot\|_1}\left(\boldsymbol{x}^k-\alpha A^\top(A\boldsymbol{x}^k-\boldsymbol{b})\right).$$

Application 2: Lasso

Consider

$$\min_{x\in\mathbb{R}^n} \quad \frac{1}{2}\|A\boldsymbol{x}-\boldsymbol{b}\|^2+\lambda\|\boldsymbol{x}\|_1,$$

where $A\in\mathbb{R}^{m\times n}$, $\boldsymbol{b}\in\mathbb{R}^m$, $\lambda>0$. The non-differentiable term is separable: $\lambda\|\boldsymbol{x}\|_1=\sum_{i=1}^{p}\lambda|x_i|$. Let $g(\boldsymbol{x}):=\frac{1}{2}\|A\boldsymbol{x}-\boldsymbol{b}\|^2$.

First, we can apply the proximal gradient (PG) method:

$$\boldsymbol{x}^{k+1}=\texttt{prox}_{\alpha\lambda\|\cdot\|_1}\left(\boldsymbol{x}^k-\alpha\nabla g(\boldsymbol{x}^k)\right)=\texttt{prox}_{\alpha\lambda\|\cdot\|_1}\left(\boldsymbol{x}^k-\alpha A^\top(A\boldsymbol{x}^k-\boldsymbol{b})\right).$$

Second, we can apply an accelerated PG (APG) or FISTA:

$$\begin{aligned} y^k &= x^k+\frac{k-1}{k+2}(x^k-x^{k-1}),\\ x^{k+1} &= \texttt{prox}_{\alpha\lambda\|\cdot\|_1}(y^k-\alpha\nabla g(y^k))\\ &= \texttt{prox}_{\alpha\lambda\|\cdot\|_1}\left(y^k-\alpha A^\top(Ay^k-\boldsymbol{b})\right). \end{aligned}$$

Application 2: Lasso (cont.)

We can also apply the coordinate descent (CD) method. Consider minimizing over x_i with all x_j, $j \neq i$ fixed. We write $\|Ax - \boldsymbol{b}\|^2$ in a quadratic form of x_i:

$$\frac{1}{2}\|A\boldsymbol{x} - \boldsymbol{b}\|^2 = \frac{1}{2}\|A_{\cdot i}x_i + \overbrace{A_{-i}\boldsymbol{x}_{-i} - \boldsymbol{b}}^{:=\Delta}\|^2$$

$$= \frac{1}{2}\|A_{\cdot i}x_i\|^2 + \langle A_{\cdot i}x_i, \Delta\rangle + \frac{\|\Delta\|^2}{2} = \frac{1}{2}\|A_{\cdot i}\|^2 x_i^2 + \left(A_{\cdot i}^\top \Delta\right) x_i + \frac{\|\Delta\|^2}{2}$$

Application 2: Lasso (cont.)

We can also apply the coordinate descent (CD) method. Consider minimizing over x_i with all x_j, $j \neq i$ fixed. We write $\|Ax - \boldsymbol{b}\|^2$ in a quadratic form of x_i:

$$\frac{1}{2}\|A\boldsymbol{x} - \boldsymbol{b}\|^2 = \frac{1}{2}\|A_{\cdot i}x_i + \overbrace{A_{-i}\boldsymbol{x}_{-i} - \boldsymbol{b}}^{:=\Delta}\|^2$$

$$= \frac{1}{2}\|A_{\cdot i}x_i\|^2 + \langle A_{\cdot i}x_i, \Delta\rangle + \frac{\|\Delta\|^2}{2} = \frac{1}{2}\|A_{\cdot i}\|^2 x_i^2 + \left(A_{\cdot i}^\top \Delta\right) x_i + \frac{\|\Delta\|^2}{2}$$

At each iteration, we solve

$$\min_{x_i} \ \frac{1}{2}\|A_{\cdot i}\|^2 x_i^2 + \left(A_{\cdot i}^\top \Delta\right) x_i + \lambda|x_i|$$

$$\overset{\text{complete squares}}{\Longleftrightarrow} \quad \min_{x_i} \ \frac{1}{2}\|A_{\cdot i}\|^2 \left(x_i + \frac{A_{\cdot i}^\top \Delta}{\|A_{\cdot i}\|^2}\right)^2 + \lambda|x_i|$$

$$\Longleftrightarrow \quad \min_{x_i} \ \frac{1}{2}\left(x_i + \frac{A_{\cdot i}^\top \Delta}{\|A_{\cdot i}\|^2}\right)^2 + \frac{\lambda}{\|A_{\cdot i}\|^2}|x_i|$$

Therefore,

$$x_i = \texttt{prox}_{\frac{\lambda}{\|A_{\cdot i}\|^2}|\cdot|}\left(-\frac{A_{\cdot i}^\top \Delta}{\|A_{\cdot i}\|^2}\right) = \texttt{prox}_{\frac{\lambda}{\|A_{\cdot i}\|^2}|\cdot|}\left(\frac{A_{\cdot i}^\top (\boldsymbol{b} - A_{-i}\boldsymbol{x}_{-i})}{\|A_{\cdot i}\|^2}\right)$$

Application 2: Lasso (cont.)

Algorithm (Coordinate Descent Method for Lasso)

Initialize x.
repeat until convergence
for $i = 1, \ldots, p$

$$x_i \quad \leftarrow \quad \texttt{prox}_{\frac{\lambda}{\|A_{\cdot i}\|^2}|\cdot|}\left(\frac{A_{\cdot i}^\top(\boldsymbol{b} - A_{-i}\boldsymbol{x}_{-i})}{\|A_{\cdot i}\|^2}\right)$$

end(for)
end(repeat)

Note: compare it with the linear regression. We here have an additional proximal operator of ℓ_1-norm, making a solution sparse.

PG vs APG vs CD

- Use the synthetic data as before
- Set $L = \lambda_{\max}(A^\top A)$, step size $\alpha = 1/L$ for PG and APG
- Plot the optimality residual

$$\left\| \boldsymbol{x}^k - \texttt{prox}_{\lambda\|\cdot\|_1}\left(\boldsymbol{x}^k - A^\top(A\boldsymbol{x}^k - \boldsymbol{b})\right) \right\|$$

  against iteration k
- Plot the first 50 entries of x.
- The estimated x from Lasso is indeed sparse.